

HiveForce Labs

THREAT ADVISORY

VULNERABILITY REPORT

D-Link Patches Wireless Routers, Fixing Multiple Critical Vulnerabilities

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A1

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TA2024358

Summary

First Seen: June 8, 2024
Affected Products: COVR-X1870, DIR-X4860, DIR-X5460

Impact: D-Link has patched several critical vulnerabilities affecting its popular wireless routers, including five flaws in total. Two of these, CVE-2024-45694 and CVE-2024-45695, are rated as critical. The remaining vulnerabilities, CVE-2024-45696, CVE-2024-45697, and CVE-2024-45698, are rated as high in severity. These issues could allow remote attackers to execute arbitrary code or gain unauthorized access through hardcoded credentials. Due to the significant security risks, users are strongly advised to update their routers immediately to mitigate these threats and prevent potential exploitation.

⚙ CVEs

CVE	NAME	AFFECTED PRODUCTS	ZERO-DAY	CISA	PATCH
CVE-2024-45694	D-link Stack-based Buffer Overflow Vulnerability	COVR-X1870, DIR-X4860, DIR-X5460	✗	✗	✓
CVE-2024-45695	D-link Stack-based Buffer Overflow Vulnerability	COVR-X1870, DIR-X4860, DIR-X5460	✗	✗	✓
CVE-2024-45696	D-link Hidden Functionality Vulnerability	COVR-X1870, DIR-X4860, DIR-X5460	✗	✗	✓
CVE-2024-45697	D-link Hidden Functionality Vulnerability	COVR-X1870, DIR-X4860, DIR-X5460	✗	✗	✓
CVE-2024-45698	D-link OS Command Injection Vulnerability	COVR-X1870, DIR-X4860, DIR-X5460	✗	✗	✓

Vulnerability Details

#1

D-Link has released crucial updates to address several security vulnerabilities found in its widely used wireless routers, including both high-end WiFi 6 models from the DIR-X series and mesh networking systems from the COVR series. These vulnerabilities pose significant risks, as they could potentially allow remote attackers to execute arbitrary code or gain unauthorized access to the devices using hardcoded credentials.

#2

The vulnerabilities identified, such as CVE-2024-45694 and CVE-2024-45695, are stack-based buffer overflow flaws within the web service of specific D-Link wireless router models. These vulnerabilities allow unauthenticated, remote attackers to exploit the flaws by sending specially crafted requests to the device, leading to arbitrary code execution. Successful exploitation could grant attackers full control over the affected routers, posing serious risks to the security of the network.

#3

CVE-2024-45696 exposes a hidden functionality that allows attackers on the local network to forcibly activate the telnet service. Once activated, they can log in using hardcoded credentials, giving them full control over the system. Similarly, CVE-2024-45697 involves a hidden functionality vulnerability that enables the telnet service when the WAN port is connected. This means a remote attacker could exploit this flaw to gain access and execute commands on the vulnerable system using hard-coded credentials.

#4

CVE-2024-45698 highlights a problem with input validation in the telnet service. Attackers can exploit this flaw by sending specially crafted input to execute arbitrary OS commands on the device. This could further compromise the device's security and functionality.

#5

D-Link has worked diligently to release the necessary patches to address these vulnerabilities. Users are strongly encouraged to upgrade to the latest firmware versions to secure their devices against these risks and prevent potential exploitation.

Vulnerabilities

CVE ID	AFFECTED PRODUCTS	AFFECTED CPE	CWE ID
CVE-2024-45694	COVR-X1870 v1.02 and below, DIR-X4860 v1.04B04_Hot-Fix and below, DIR-X5460 v1.11B01_Hot-Fix and below	cpe:2.3:h:d-link:dir-x5460:*:*:*:*:*:*	CWE-121
CVE-2024-45695			CWE-121
CVE-2024-45696		cpe:2.3:h:d-link:covr-X1870:*:*:*:*:*:*	CWE-912
CVE-2024-45697		cpe:2.3:h:d-link:dir-x4860:*:*:*:*:*:*	CWE-912
CVE-2024-45698			CWE-78

Recommendations



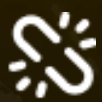
Update: All users are urged to install the necessary updates and regularly check for additional updates to maintain security. After downloading the update, it is critical to verify the update’s success by comparing the software version displayed in the router’s interface with the version provided in the update. This ensures the system is fully protected and running the latest security patches.



Regularly Monitor Device Logs: Regularly check your router’s logs for any unusual activity or unauthorized access attempts. Monitoring logs can help you detect potential security breaches early.



Disable Unused Services: Ensure that the Telnet service is disabled, particularly on any device exposed to the internet via WAN. Use more secure protocols such as SSH for remote access. If Telnet must be enabled for legacy reasons, restrict access to only trusted IPs or within a secured network perimeter.



Vulnerability Management: Implement a robust vulnerability management process to ensure that software and systems are regularly assessed for vulnerabilities and updated with the required security patches. Prioritize critical vulnerabilities identified by security advisories and vendors to mitigate the risk of exploitation by threat actors.

Potential MITRE ATT&CK TTPs

<u>TA0043</u> Reconnaissance	<u>TA0042</u> Resource Development	<u>TA0001</u> Initial Access	<u>TA0002</u> Execution
<u>TA0008</u> Lateral Movement	<u>T1588</u> Obtain Capabilities	<u>T1588.006</u> Vulnerabilities	<u>T1059</u> Command and Scripting Interpreter
<u>T1021</u> Remote Services	<u>T1078</u> Valid Accounts	<u>T1203</u> Exploitation for Client Execution	<u>T1589</u> Gather Victim Identity Information
<u>T1589.001</u> Credentials			

Patch Details

All users of D-Link are urged to promptly install the necessary updates and to routinely check for future updates to ensure continued security. Upgrade and install the patched version as follows:

COVR-X1870 - v1.03B01

DIR-X4860 - v1.04B05

DIR-X5460 - DIR-X5460A1_V1.11B04

Link:

<https://supportannouncement.us.dlink.com/security/publication.aspx?name=SAP10412>

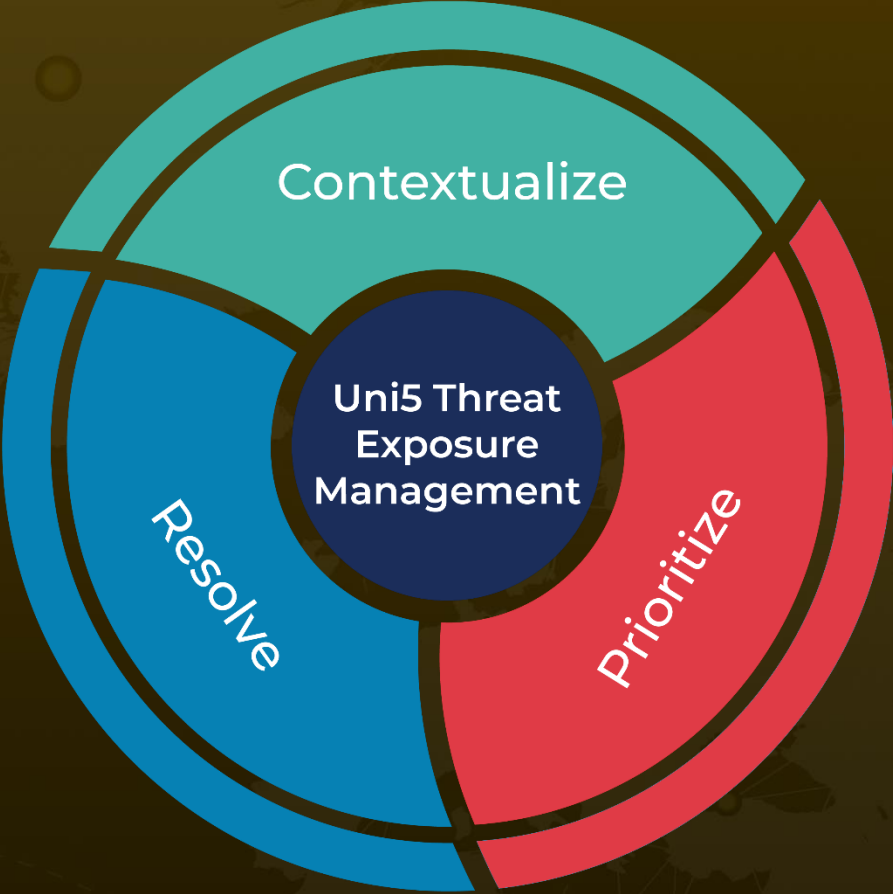
References

<https://supportannouncement.us.dlink.com/security/publication.aspx?name=SAP10412>

What Next?

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